1.29 Binomial Expansion

Chapter 8 · Original slide 4

Starter

4

a) Expand $(a+b)^0$
b) Expand $(a+b)^1$
c) Expand $(a+b)^2$
d) Expand $(a+b)^3$
e) Expand $(a+b)^4$

What do you notice about:

The coefficients: They follow Pascal's triangle (we'll explore on next slide).
The powers of a and b: Power of a decreases each time (starting at the power)
Power of b increases each time (starting at 0)

4

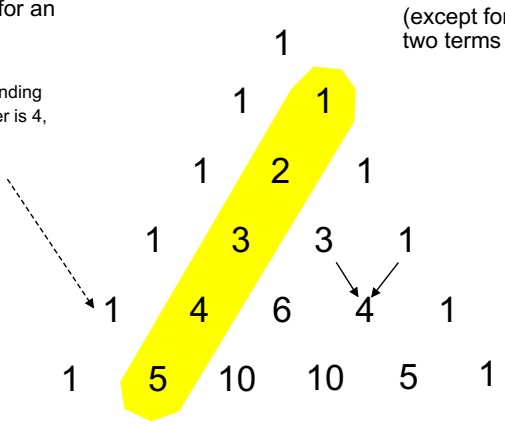
Chapter 8 · Original slide 5

More on Pascal's Triangle

5

The second number of each row tells us what row we should use for an expansion.

So if we were expanding $(2+x)^4$, the power is 4, so we use this row.



In Pascal's Triangle, each term (except for the 1s) is the sum of the two terms above.

Tip: memorise each row up to what you see here.

We'll see later WHY each row gives us the coefficients in an expansion of $(a+b)^n$

5

1.29 Binomial Expansion

Chapter 8 · Original slide 6

Example

6

Find the expansion of $(2+3x)^4$

Next have descending or ascending powers of one of the terms, going between 0 and 4 (note that if the power is 0, the term is 1, so we need not write it).

$$(2+3x)^4 =$$

First fill in the correct row of Pascal's triangle.

And do the same with the second term but with powers going the opposite way, noting again that the 'power of 0' term does not appear.

Simplify each term (ensuring any number in a bracket is raised to the appropriate power)

Tip: Initially write *one line per term* for your expansion (before you simplify at the end), as we have done above. There will be less faffing trying to ensure you have enough space for each term.

6

Chapter 8 · Original slide 7

Another Example

7

$(1-2x)$ is the same as $(1+(-2x))$, so we expand as before, but use $-2x$ for the second term.

$$(1-2x)^3 =$$

Tip: If one of the terms in the original bracket is negative, the terms in your expansion will oscillate between positive and negative. If they don't (e.g. two consecutive negatives), you've done something wrong!

7

1.29 Binomial Expansion

Chapter 8 · Original slide 8

Getting a single term in the expansion

8

The coefficient of x^2 in the expansion of $(2 - cx)^5$ is 720.
Find the possible value(s) of the constant c.

8

Chapter 8 · Original slide 9

Test Your Understanding

9

Edexcel C2

(a) Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(2 + kx)^7$
where k is a constant. Give each term in its simplest form. \hfill(4)
Given that the coefficient of x^2 is 6 times the coefficient of x,
(b) find the value of k. \hfill(2)

(a)

(b)

9

1.29 Binomial Expansion

Chapter 8 · Original slide 10

Exercise 8A – Y1 page 160-161

4 Fully expand the expression $(1+3x)(1+2x)^3$.

5 Expand $(2+y)^3$. Hence or otherwise, write down the expansion of $(2+x-x^2)^3$ in ascending powers of x.

Chapter 8 · Original slide 11

Exercise 8A – Y1 page 160-161

6 The coefficient of x^2 in the expansion of $(2+ax)^3$ is 54. Find the possible values of the constant a.

7 The coefficient of x^3 in the expansion of $(2-x)(3+bx)^3$ is 45. Find possible values of the constant b.

1.29 Binomial Expansion

Chapter 8 · Original slide 12

Exercise 8A – Y1 page 160-161

12

8 Work out the coefficient of x^2 in the expansion of $(p-2x)^3$. Give your answer in terms of p.

9 After 5 years, the value of an investment of \pounds500 at an interest rate of $X\%$ per annum is given by $500\left(1+\frac{X}{100}\right)^5$

Find an approximation for this expression in the form $A+BX+CX^2$, where A, B and C are constants to be found. You can ignore higher powers of X.

12

Chapter 8 · Original slide 13

Extension

13

[MAT 2009 1J]

The number of pairs of positive integers x, y which solve the equation:

$$x^3+6x^2y+12xy^2+8y^3=2^{30}$$

is:

A) 0
B) 2^6
C) 2^9-1
D) $2^{10}+2$

13

1.29 Binomial Expansion

Chapter 8 · Original slide 14

Factorial and Choose Function

14

$n! = n \times (n-1) \times (n-2) \times \ldots \times 2 \times 1$

said "n factorial", is the number of ways of arranging n objects in a line.

For example, suppose you had three letters, A, B and C, and wanted to arrange them in a line to form a 'word', e.g. ACB or BAC.

- There are 3 choices for the first letter.
- There are then 2 choices left for the second letter.
- There is then only 1 choice left for the last letter.
-

There are therefore $3 \times 2 \times 1 = 3! = 6$ possible combinations.

Your calculator can calculate a factorial using the $\mathbf{x!}$ button.

$nCr = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

said "n choose r", is the number of ways of 'choosing' r things from n, such that the order in our selection does not matter.
These are also known as **binomial coefficients**.

For example, if you a football team captain and need to choose 4 people from amongst 10 in your class, there are $\binom{10}{4} = \frac{10!}{4!6!} = 210$ possible selections.

(Note: the $\binom{10}{4}$ notation is preferable to $10C4$)

Use the $\mathbf{nCr}$ button on your calculator (your calculator input should display "10C4")

14

Chapter 8 · Original slide 15

Examples

15

Calculate the value of the following. You may use the factorial button, but not the nCr button.

a) $5!$

b) $\binom{5}{3}$

c) $0!$

d) $\binom{20}{1}$

e) $\binom{20}{0}$

- $0! = 1$. Accept this for the moment, but all will be explained in part (e).
-
- ! $\binom{n}{1} = n$ for all n.
- ! $\binom{n}{0} = 1$.
  $\binom{n}{2} = \frac{n(n-1)}{2}$ for all n.
- In general, where the bottom number is above half of the top, we can subtract it from the top, i.e. $\binom{20}{18} = \binom{20}{2}$.
-
-
-

15

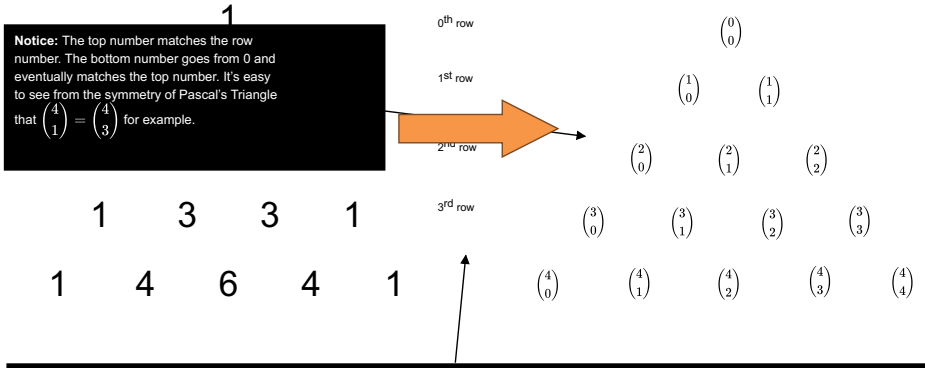
1.29 Binomial Expansion

Chapter 8 · Original slide 16

Why do we care?

16

If the power in the binomial expansion is large, e.g. $(x+3)^{20}$, it is no longer practical to go this far down Pascal's triangle. We can instead use the choose function to get numbers from anywhere within the triangle. We'll practise doing this after the next exercise.



Notice: The top number matches the row number. The bottom number goes from 0 and eventually matches the top number. It's easy to see from the symmetry of Pascal's Triangle that $\binom{4}{1}=\binom{4}{3}$ for example.

Textbook Note: The textbook refers to the top row as the "1st row" and the first number in each row as the "1st entry". This might sound sensible, but is against accepted practice: It makes much more sense that the row number matches the number at the top of the binomial coefficient, and the entry number matches the bottom number. We therefore call the top row the "0th row" and the first entry of each row the "0th entry". So the kth entry of the nth row of Pascal's Triangle is therefore a nice clean $\binom{n}{k}$, not $\binom{n-1}{k-1}$ as suggested by the textbook.

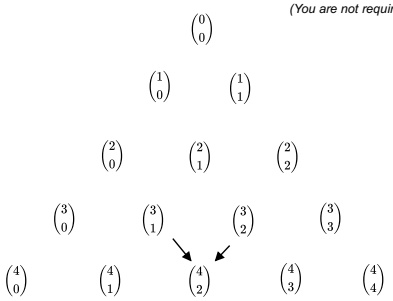
16

Chapter 8 · Original slide 17

Interesting facts

17

(You are not required to know this, but it is helpful for STEP)



We earlier saw that each entry of Pascal's Triangle is the sum of the two above it. Thus for example:

$$\binom{3}{1}+\binom{3}{2}=\binom{4}{2}$$

More generally:

$$\binom{n-1}{k-1}+\binom{n-1}{k}=\binom{n}{k}$$

This is known as Pascal's Rule.

Informal proof of Pascal's Rule:

Suppose I have n items and I have to choose k of them. Clearly there's $\binom{n}{k}$ possible selections. But we could also find the number of selections by considering the first item of the n available:

- It might be chosen. If so, we have $k-1$ items left to choose from amongst the $n-1$ remaining. That's $\binom{n-1}{k-1}$ possible selections.
- Otherwise it is not chosen. We still have k items to choose, from amongst the remaining $n-1$ items. That's $\binom{n-1}{k}$ possible selections.

Thus in total there are $\binom{n-1}{k-1}+\binom{n-1}{k}$ possible selections.

17

1.29 Binomial Expansion

Chapter 8 · Original slide 18

Exercise 8B – Y1 page 162

18

10 Given that $\binom{50}{13} = \frac{50!}{13!a!}$, write down the value of a.

11 Given that $\binom{35}{p} = \frac{35!}{p!18!}$, write down the value of p.

18

Chapter 8 · Original slide 19

Exercise 8B – Y1 page 162

19

Challenge
a Work out ${}^{10}C_3$ and ${}^{10}C_7$.
b Work out ${}^{14}C_5$ and ${}^{14}C_9$.
c What do you notice about your answers to parts **a** and **b**?
d Prove that ${}^{n}C_r = {}^{n}C_{n-r}$.

19

1.29 Binomial Expansion

Chapter 8 · Original slide 20

Using Binomial Coefficients to Expand

20

In the previous section we learnt about the ‘choose’ function and how this related to Pascal’s Triangle.

1 5 10 10 5 1

$$\binom{5}{0}\ \binom{5}{1}\ \binom{5}{2}\ \binom{5}{3}\ \binom{5}{4}\ \binom{5}{5}$$

Why do rows of Pascal’s Triangle give us the coefficients in a Binomial Expansion?

Consider: $(a+b)^5 = (a+b)(a+b)(a+b)(a+b)(a+b)$
Each term of the expansion involves picking one term from each bracket.
How many times will a^3b^2 appear in the expansion?

20

Chapter 8 · Original slide 21

Using Binomial Coefficients to Expand

21

$\mathbb{N}$ is the set of natural numbers, i.e. positive integers. This formula is only valid for positive integers n. In Year 2 you will see how to deal with fractional/negative n.

The binomial expansion, when $n \in \mathbb{N}$:

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{r}a^{n-r}b^r + \ldots + b^n$$

Find the first 4 terms in the expansion of $(3x+1)^{10}$, in ascending powers of x.

$$(3x+1)^{10} =$$

This is exactly the same method as before, except we’ve just had to calculate the Binomial coefficients ourselves rather than read them off Pascal’s Triangle.

21

1.29 Binomial Expansion

Chapter 8 · Original slide 22

Test Your Understanding

22

Find the first 3 terms in the expansion of $\left(2-\frac{1}{3}x\right)^7$, in ascending powers of x.

Note: The "$+\ldots$" indicates that there would have been other terms in the expansion.

22

Chapter 8 · Original slide 23

Exercise 8C – Y1 page 164

23

5 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(2-x)^6$ and simplify each term.

6 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(3-2x)^5$ giving each term in its simplest form.

7 Find the binomial expansion of $\left(x+\frac{1}{x}\right)^5$ giving each term in its simplest form.

23

1.29 Binomial Expansion

Chapter 8 · Original slide 24

Extension

24

1 [AEA 2013 Q1a] In the binomial expansion of $\left(1+\frac{12n}{5}x\right)^n$ the coefficients of x^2 and x^3 are equal and non-zero. Find the possible values of n.

Hint: Remember that $\binom{n}{2} = \frac{n(n-1)}{2!}$

Can you similarly simplify $\binom{n}{3}$ using $\binom{n}{r} = \frac{n!}{r!(n-r)!}$?

2 [STEP I 2010 Q5a] By considering the expansion of $(1+x)^n$, where n is a positive integer, or otherwise, show that:
$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \ldots + \binom{n}{n} = 2^n$

reflection: This means that the sum of each row in Pascal's Triangle gives successive powers of 2. Safe!

24

Chapter 8 · Original slide 25

Getting a single term in the expansion

25

In the expansion of $(a+b)^n$ the general term is given by $\binom{n}{r}a^{n-r}b^r$

Expression	Power of x in term wanted.	Term in expansion
$(a+x)^{10}$	3	
$(2x-1)^{75}$	50	
$(3-x)^{12}$	7	
$(3x+4)^{16}$	3	

Note: The two powers add up to 10.

25

1.29 Binomial Expansion

Chapter 8 · Original slide 26

Getting a single term in the expansion

26

The coefficient of x^4 in the expansion of $(1+qx)^{10}$ is 3360.
Find the possible value(s) of the constant q.

26

Chapter 8 · Original slide 27

Test Your Understanding

27

In the expansion of $(1+ax)^{10}$, where a is a non-zero constant the coefficient of x^3 is double the coefficient of x^2.
Find the value of a.

27

1.29 Binomial Expansion

Chapter 8 · Original slide 28

Exercise 8D – Y1 page 166-167

28

5 When $(1-2x)^p$ is expanded, the coefficient of x^2 is 40. Given that $p > 0$, use this information to find:

a the value of the constant p
b the coefficient of x
c the coefficient of x^3

28

Chapter 8 · Original slide 29

Exercise 8D – Y1 page 166-167

29

6 a Find the first three terms, in ascending powers of x, of the binomial expansion of $(5+px)^{30}$, where p is a non-zero constant.
b Given that in this expansion the coefficient of x^2 is 29 times the coefficient of x work out the value of p.

29

1.29 Binomial Expansion

Chapter 8 · Original slide 30

Exercise 8D – Y1 page 166-167

30

7 a Find the first four terms, in ascending powers of x, of the binomial expansion of $(1 + qx)^{10}$, where q is a non-zero constant.
b Given that in the expansion of $(1 + qx)^{10}$ the coefficient of x^3 is 108 times the coefficient of x, work out the value of q.

30

Chapter 8 · Original slide 31

Exercise 8D – Y1 page 166-167

31

8 a Find the first three terms, in ascending powers of x, of the binomial expansion of $(1 + px)^{11}$, where p is a constant.
b The first 3 terms in the same expansion are $1, 77x$ and qx^2, where q is a constant. Find the value of p and the value of q.

31

1.29 Binomial Expansion

Chapter 8 · Original slide 32

Exercise 8D – Y1 page 166-167

32

9 a Write down the first three terms, in ascending powers of x, of the binomial expansion of $(1+px)^{15}$, where p is a non-zero constant.
b Given that, in the expansion of $(1+px)^{15}$, the coefficient of x is $(-q)$ and the coefficient of x^2 is $5q$, find the value of p and the value of q.

32

Chapter 8 · Original slide 33

Exercise 8D – Y1 page 166-167

33

10 In the binomial expansion of $(1+x)^{30}$, the coefficients of x^9 and x^{10} are p and q respectively. Find the value of $\frac{q}{p}$.
\hfill$(4\text{ mar}\ldots$

33

1.29 Binomial Expansion

Chapter 8 · Original slide 34

Extension

34

1 *[MAT 2014 1G]* Let n be a positive integer. The coefficient of x^3y^5 in the expansion of $\left(1+xy+y^2\right)^n$ equals:

A) n

B) 2^n

C) $\binom{n}{3}\binom{n}{5}$

D) $4\binom{n}{4}$

E) $\binom{n}{8}$

2 [STEP I 2013 Q6] By considering the coefficient of x^r in the series for $(1+x)(1+x)^n$, or otherwise, obtain the following relation between binomial coefficients:
$\binom{n}{r}+\binom{n}{r-1}=\binom{n+1}{r}$

34

Chapter 8 · Original slide 35

Estimating Powers

35

Edexcel C2 Jan 2012 Q3

(a) Find the first 4 terms of the binomial expansion, in ascending powers of x, of $\left(1+\frac{x}{4}\right)^8$, giving each term in its simplest form. \hfill(4)
(b) Use your expansion to estimate the value of $(1.025)^8$, giving your answer to 4 decimal places. \hfill(3)

Tip: Use your calculator to compare against the exact value of 1.025^8.

a

b Why should this be a reasonably good approximation of 1.025^8 despite the missing terms in the expansion?

35

1.29 Binomial Expansion

Chapter 8 · Original slide 36

Test Your Understanding

36

Edexcel C2 Jan 2008 Q3

(a) Find the first 4 terms of the expansion of $\left(1+\frac{x}{2}\right)^{10}$ in ascending powers of x, giving each term in its simplest form. \hfill(4)
(b) Use your expansion to estimate the value of $(1.005)^{10}$, giving your answer to 5 decimal places. \hfill(3)

(a)

(b)

36

Chapter 8 · Original slide 37

Exercise 8E – Y1 Page 168-169

37

7 a Find the first 3 terms, in ascending powers of x, of the binomial expansion of $\left(3-\frac{x}{5}\right)^{10}$, giving each term in its simplest form. \hfill$(4\text{ m}\ldots$
b Explain how you would use your expansion to give an estimate for the value of 2.98^{10}. \hfill$(1\ldots$

37

1.29 Binomial Expansion

Chapter 8 · Original slide 38

Exercise 8E – Y1 Page 168-169

38

8 a Find the first 4 terms, in ascending powers of x, of the binomial expansion of $(1-3x)^5$. Give each term in its simplest form. \hfill$(4\ldots$

b If x is small, so that x^2 and higher powers can be ignored, show that $(1+x)(1-3x)^5 \approx 1-14x$. \hfill$(2$

38

Chapter 8 · Original slide 39

Exercise 8E – Y1 Page 168-169

39

9 A microchip company models the probability of having no faulty chips on a single production run as:

$\mathrm{P}(\text{no fault}) = (1-p)^n,\ p < 0.001$

where p is the probability of a single chip being faulty, and n being the total number of chips produced.

a State why the model is restricted to small values of p. \hfill$(1\ \text{mar}\ldots$

b Given that $n = 200$, find an approximate expression for $\mathrm{P}(\text{no fault})$ in the form $a + bp + cp^2$. \hfill $(2\ \text{mark}\ldots$

c The company wants to achieve a 92% likelihood of having no faulty chips on a production run of 200 chips. Use your answer to part **b** to suggest a maximum value of p for this to be the case. \hfill$(4\ \text{m}$

39